

Relationship Between Food Waste Abundance and Flight Initiation Distance of Urban Crows at University of British Columbia

INTRODUCTION

Urban environments are home to many animals who live besides humans. These animals include crows, squirrels, racoons, cats, etc. These animals are referred to as "urban animals." They can sometimes be found in high abundance in areas with a large amount of human-caused food waste, as it is a common food resource for them. This waste is much more common in some areas, like outside restaurants, near garbage bins, and at picnic spots, compared to others. These are very accessible to wildlife, and some species like crows are seen exploiting this.

Human waste has also been linked to changes in animal behaviours (Newsome et al 2017). Corvid birds such as crows have shown to have high adaptability for human environments (Benmazou et al 2021). The availability of waste can play a major role when looking at animal behaviors as in areas with high waste, animals might be more tolerant to humans due to their daily interactions and threats compared to areas where food waste is not common; they might be more scared as they do not interact with them as much. This change in behaviour can influence how animals interact with their surrounding environments.

This project investigates the change in behavior in *Corvus brachyrhynchos* (crows) as the abundance of waste increases. I used Flight Initiation Distance (FID) to find whether crows feel more threatened in areas with high to low food abundance. My hypothesis is that crows located in areas with higher waste abundance will have shorter FID because they are more exposed to humans and have greater food resources.

METHODS

I observed different locations around the University of British Columbia, which were selected based on waste and crow abundances. These include Main Mall, University Boulevard, Memorial Road, etc. Food waste abundance was quantified by counting the number of food waste items found in a 2x2 or 4m² square, and only compost items were counted.

Flight Initiation Distance was measured by approaching crows directly in a straight path at a steady pace. All observations came from crows, which were already on the ground, and no crows were baited or given food. While walking straight towards the crow, which was sitting, the distance from me and the crow, where it first moved or flew away due to me, was recorded. In total, 22 crows were observed in different locations across campus. All observations followed the same procedure.

RESULTS

Table 1. Raw observations of crows (species), Flight Initiation Distance (FID), and food waste abundance (Abundance) collected at different locations across University of British Columbia (Site) on June 17th, 2026 (n=22). Food waste abundance represents the number of food waste items seen in a 4m² sampling area where the crow was spotted.

Site	area (m ²)	Abundance	species	FID (m)
University blvd loaf	4m	10	Crow	2.18
University blvd Swiss bakery	4m	7	Crow	1.59
Jim Everett memorial park	4m	22	Crow	1.37
Jim Everett memorial park	4m	22	Crow	1.63
Jim Everett memorial park	4m	25	Crow	1.14
Jim Everett memorial park	4m	14	Crow	2.1
Jim Everett memorial park	4m	23	Crow	1.52
Jim Everett memorial park	4m	24	Crow	1.54
Jim Everett memorial park	4m	21	Crow	1.7
Jim Everett memorial park	4m	12	Crow	2.22
Jim Everett memorial park	4m	19	Crow	1.56
Jim Everett memorial park	4m	18	Crow	1.89
University blvd McDonalds	4m	16	Crow	1.98
University blvd kinesiology building	4m	4	Crow	2.79
Main mall flag post	4m	2	Crow	2.45
Main mall flag post	4m	1	Crow	3.11
Main mall flag post	4m	2	Crow	2.94
Memorial road	4m	0	Crow	3.14
Triple O	4m	6	Crow	1.32
Triple O	4m	5	Crow	1.39
Main mall Earth science building	4m	1	Crow	2.84
Main mall Earth science building	4m	0	Crow	3.79

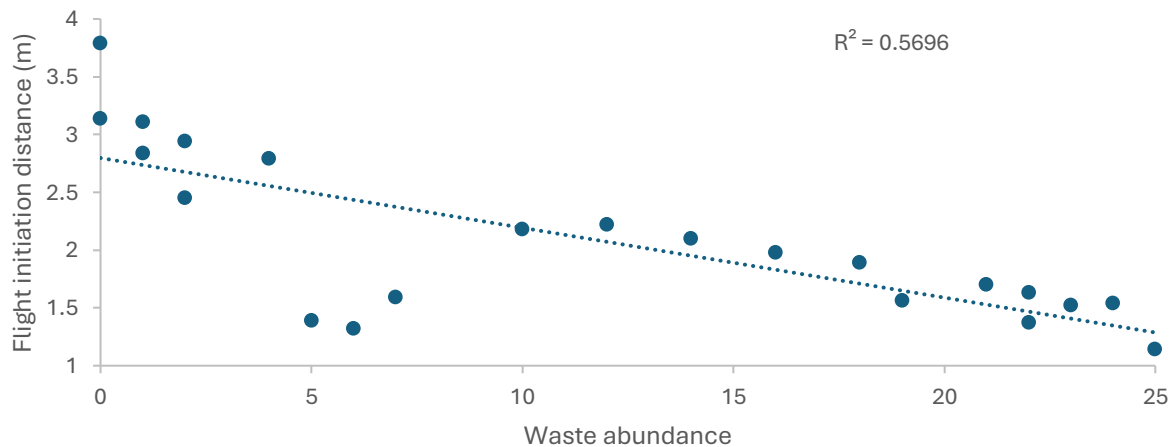


Figure 1. Flight initiation distance (meters) was negatively related to food waste abundance. ($R^2=0.57$, $r=-0.75$, $N=22$). Data was collected at multiple sites at the University of British Columbia in June, 2026.

In total, 22 observations of crows were collected across 9 different locations on campus (Table 1). Food waste abundance ranged from 0 to 25, which was seen in a 4m² sampling area, and FID ranged from 1.14 m to 3.79 m. Measurements were collected using a tape measure.

A negative relationship between food waste abundance and FID was observed (Figure 1). Crows observed in locations with higher amounts of food waste generally have shorter FID compared to crows with little or no waste. The p value of 0.00004 suggests that this relationship is not due to chance.

The results support the hypothesis that crows located in areas with greater food abundance would have shorter FID. The negative relationship shows that crows in higher food waste areas allow people to get more closer to them.

However, due to time, there are several limitations that should be considered.

1. The sample size is relatively small, and there were only select locations where crows were seen.
2. Although I tried my best not to observe the same crow again, there is no way to differentiate that the same crow might have been observed more than once.
3. Due to not having adequate equipment, the sample location of 4m² is very small, and a bigger size should be considered.
4. As all data was collected in one day, a long-term pattern of behavior change might not be feasible to conclude
5. Other factors such as human activity levels and distance from nearest bin or waste etc. could play an effect that was not captured

In conclusion, there does seem to be a negative relationship between food waste abundance and FID in urban crows across UBC. However, due to the scope and limitations of the project, multiple different limitations and other mechanisms could explain this as behavioural change as well.

Works referenced

https://portal.gis.ubc.ca/arcgis/apps/experiencebuilder/experience/?data_id=dataSource_11-49d854b6077949b48d2da5625612e8eb%3A1&id=80b5c05856fe4e46a3c1047e4d27f3bb&page=page_7

- Used to check for abundance of animals across campus

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